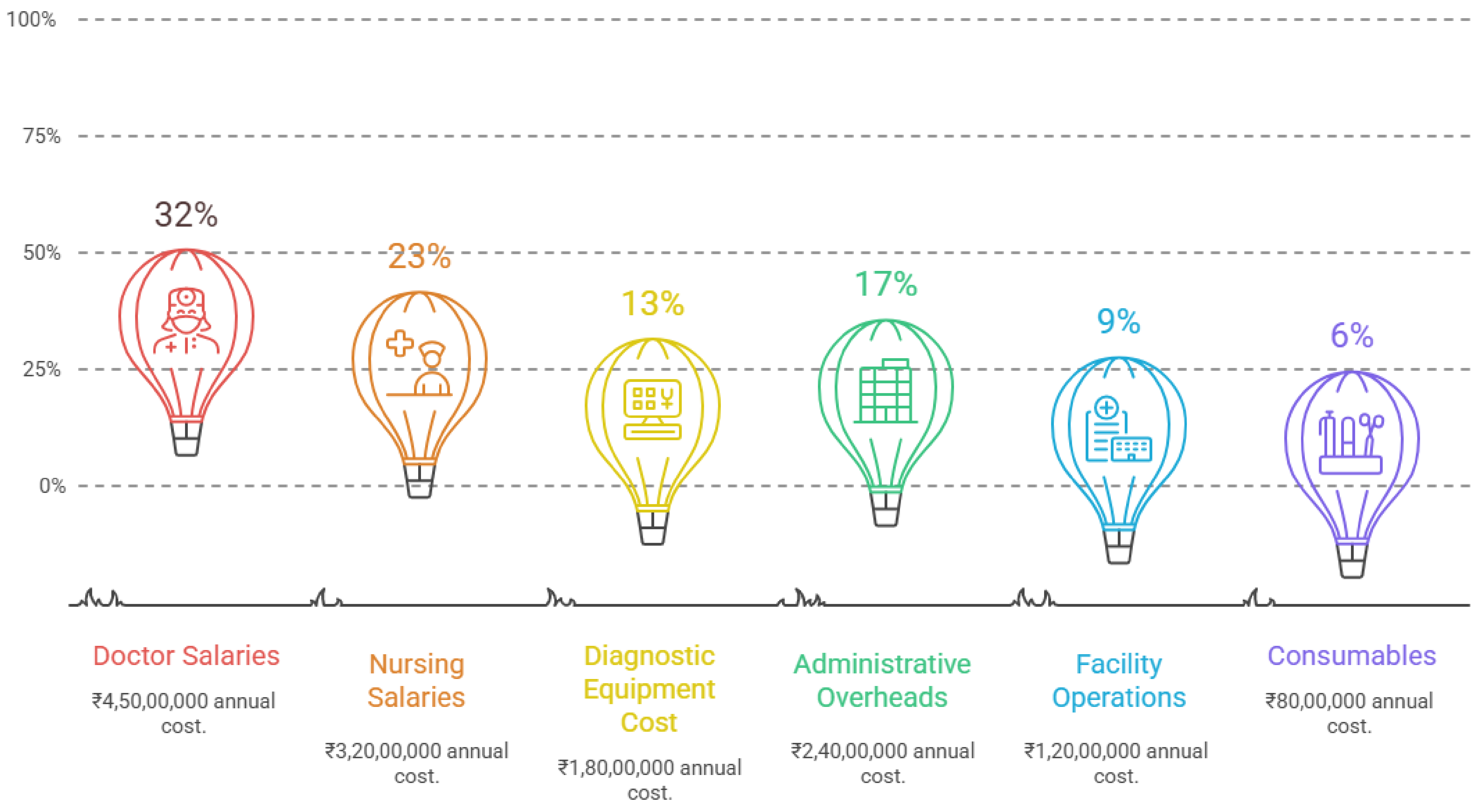
Activity - 1

Objective - Understanding True Hospital Service Costs Using Activity-Based Costing



Hospital Anuual Cost Summary

Activity	Description
Activity A	Outpatient Consultation
Activity B	Diagnostic Imaging (X-ray/CT)
Activity C	Inpatient Admission (General Ward)

Activities Analyzed

Activity Volumes & Cost Drivers

Activity	Annual Volume	Cost Driver
Outpatient Consultations	60,000 visits	No. of consultations
Diagnostic Imaging	18,000 scans	Machine hours
Inpatient Admissions	6,000 admissions	Patient days

Cost Driver Distribution

Cost Component	Outpatient	Diagnostic	Inpatient
Doctor Time	40%	20%	40%
Nursing Time	10%	20%	70%
Equipment Use	0%	100%	0%
Admin Overheads	50%	20%	30%
Facility Operations	20%	20%	60%
Consumables	30%	30%	40%

Calculation Example-

Doctor Salaries (₹4,50,00,000)

- Outpatient (40%): ₹1,80,00,000
- Diagnostics (20%): ₹90,00,000
- Inpatient (40%): ₹1,80,00,000

ABC Cost Allocation

Cost Component	Doctor Salaries	Nursing Salaries	Diagnostic Equipment	Admin Overheads	Facility Operations	Consumables	Total Cost per Activity
Total Cost (₹)	4,50,00,000	3,20,00,000	1,90,00,000	2,40,00,000	1,20,00,000	80,00,000	14,00,00,000
Outpatient (₹)	1,80,00,000	32,00,000	0	1,20,00,000	24,00,000	24,00,000	3,80,00,000
Diagnostics (₹)	90,00,000	64,00,000	1,90,00,000	48,00,000	24,00,000	24,00,000	4,40,00,000
Inpatient (₹)	1,80,00,000	2,24,00,000	0	72,00,000	72,00,000	32,00,000	5,80,00,000

Total Cost per Activity

Outpatient Consultation – Total Cost

- Doctor: 4.5 Cr × 40% = 1.80 Cr
- Nursing: 3.2 Cr × 10% = 0.32 Cr
- Admin: 2.4 Cr × 50% = 1.20 Cr
- Facility: 1.2 Cr × 20% = 0.24 Cr
- Consumables: 0.8 Cr × 30% = 0.24 Cr

Total Outpatient Cost = ₹3.80 Cr

Diagnostic Imaging – Total Cost

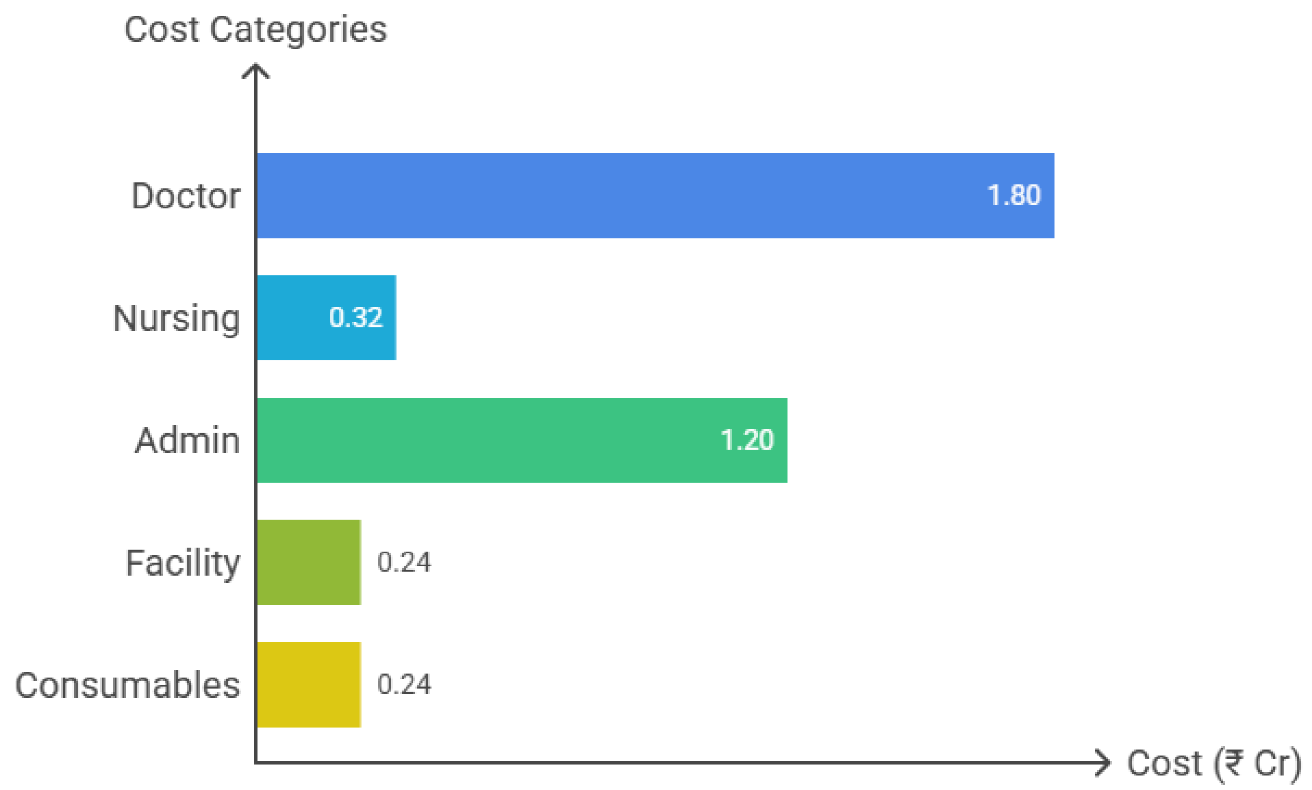
- Doctor: 4.5 Cr × 20% = 0.90 Cr
- Nursing: 3.2 Cr × 20% = 0.64 Cr
- Equipment: 1.8 Cr × 100% = 1.80 Cr
- Admin: 2.4 Cr × 20% = 0.48 Cr
- Facility: 1.2 Cr × 20% = 0.24 Cr
- Consumables: 0.8 Cr × 30% = 0.24 Cr

Total Diagnostic Cost = ₹4.30 Cr

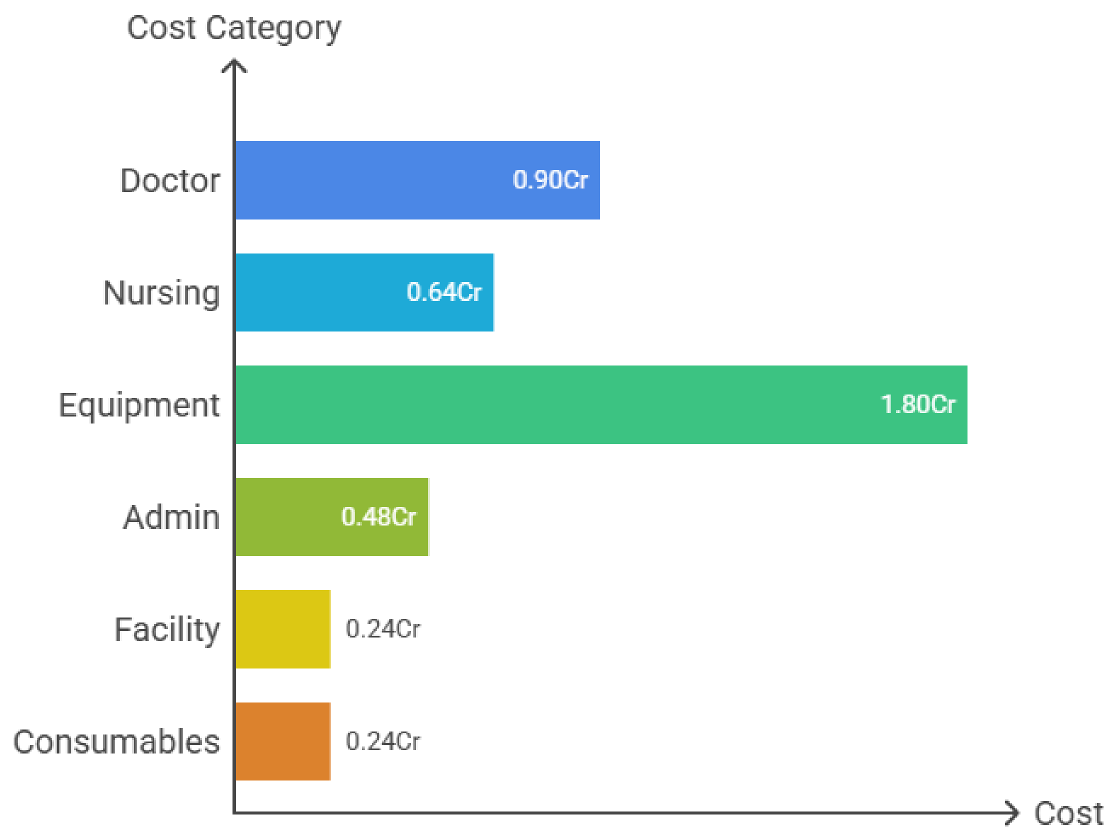
Inpatient Admission – Total Cost

- Doctor: 4.5 Cr × 40% = 1.80 Cr
- Nursing: 3.2 Cr × 70% = 2.24 Cr
- Admin: 2.4 Cr × 30% = 0.72 Cr
- Facility: 1.2 Cr × 60% = 0.72 Cr
- Consumables: 0.8 Cr × 40% = 0.32 Cr

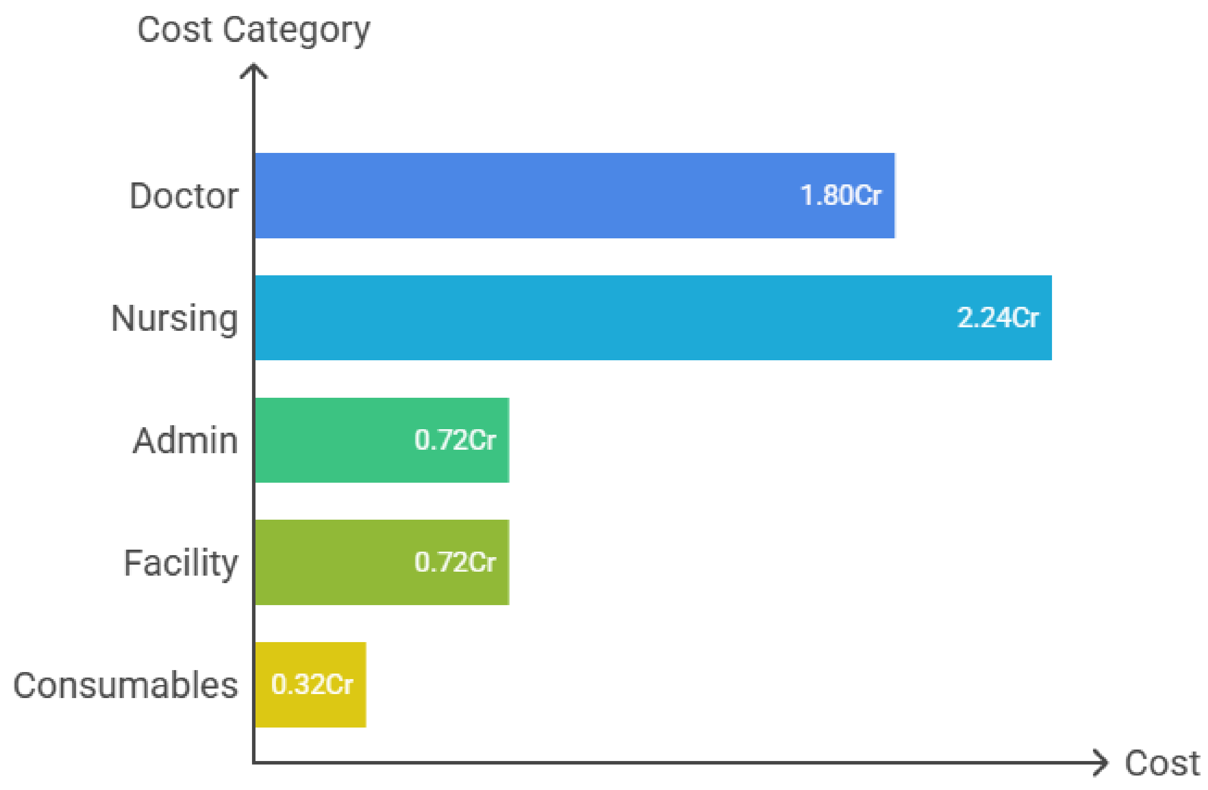
Total Inpatient Cost = ₹5.80 Cr



Cost Breakdown for Outpatient Consultation



Cost Breakdown for Diagnostic Imaging



Cost Breakdown for Inpatient Services

Cost per Unit by Activity

₹633

Outpatient Consultation Cost per Unit

Total cost is ₹3.80 Cr for 60,000 visits.

₹2,389

Diagnostic Imaging Cost per Unit

Total cost is ₹4.30 Cr for 18,000 scans.

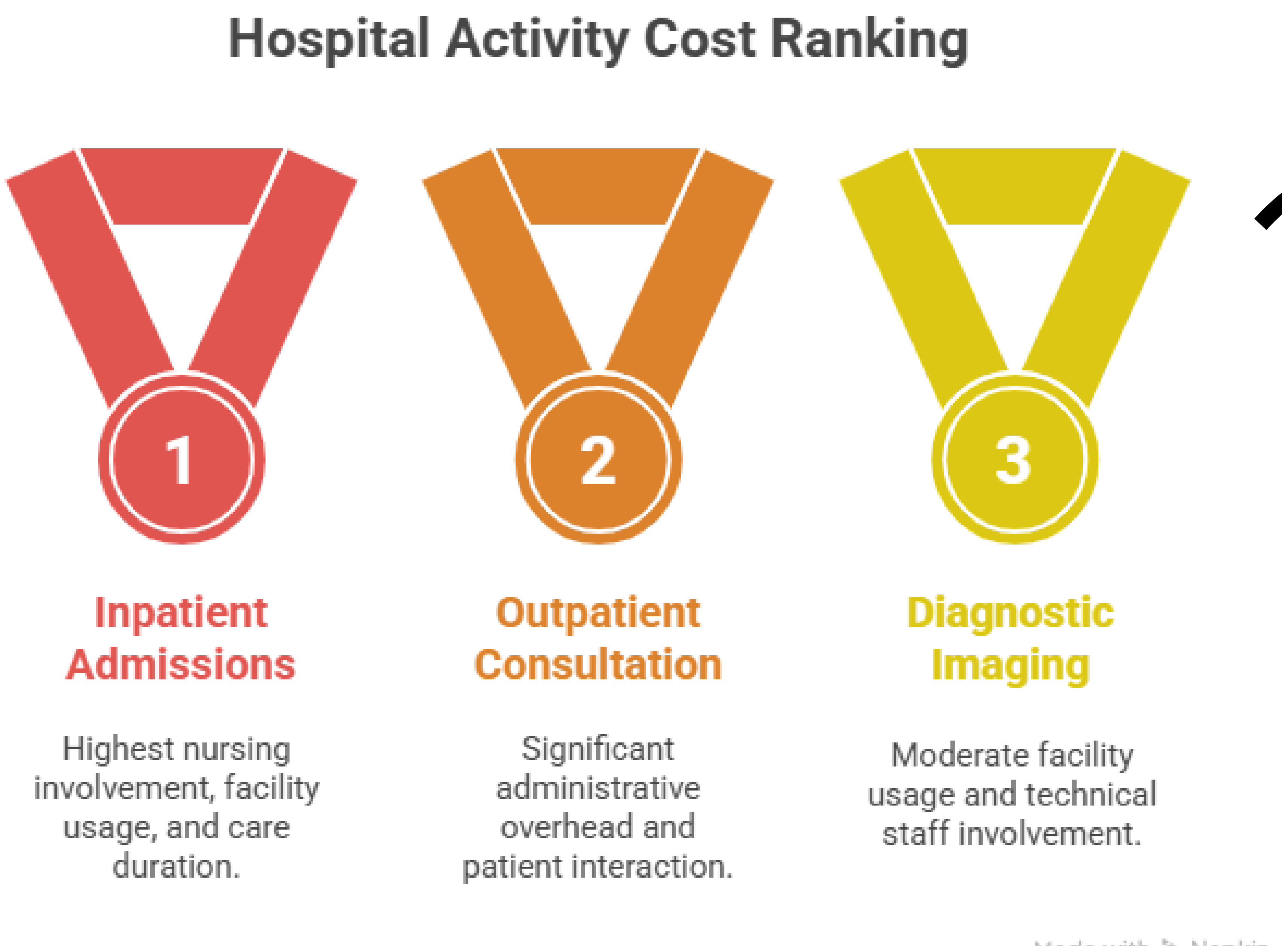
₹9,667

Inpatient Admission Cost per Unit

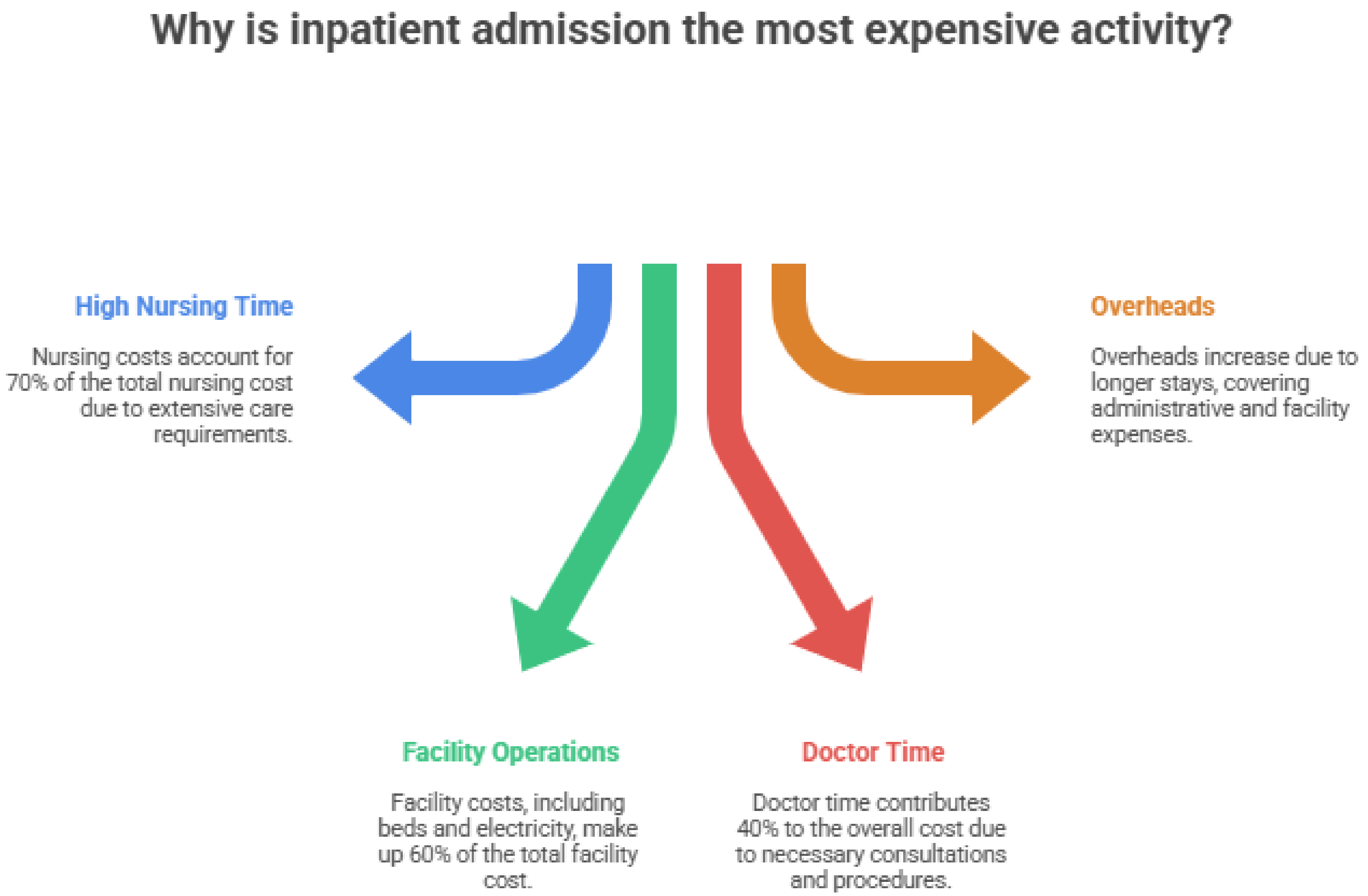
Total cost is ₹5.80 Cr for 6,000 admissions.

Inpatient admission is the most expensive activity per unit.

Analysis



Hospital Activity Cost Ranking
1
2
3
Inpatient Admissions
Highest nursing involvement, facility usage, and care duration.
Outpatient Consultation
Significant administrative overhead and patient interaction.
Diagnostic Imaging
Moderate facility usage and technical staff involvement.



Why is inpatient admission the most expensive activity?
High Nursing Time
Nursing costs account for 70% of the total nursing cost due to extensive care requirements.
Overheads
Overheads increase due to longer stays, covering administrative and facility expenses.
Facility Operations
Facility costs, including beds and electricity, make up 60% of the total facility cost.
Doctor Time
Doctor time contributes 40% to the overall cost due to necessary consultations and procedures.

Which costing method should be used to calculate hospital activity costs?

Traditional vs. ABC Costing

Characteristic	Indirect Cost Allocation	Cost Accuracy	Example
Traditional Costing	Evenly across all services	Less accurate	Splitting expenses equally
ABC Costing	Based on resource usage	More accurate	Checking actual activity costs

Overhead Absorption in Hospitals

Which activity shows the greatest overhead absorption?

Inpatient Admissions. They take the largest share of nursing salaries, facility operations, and admin overheads.

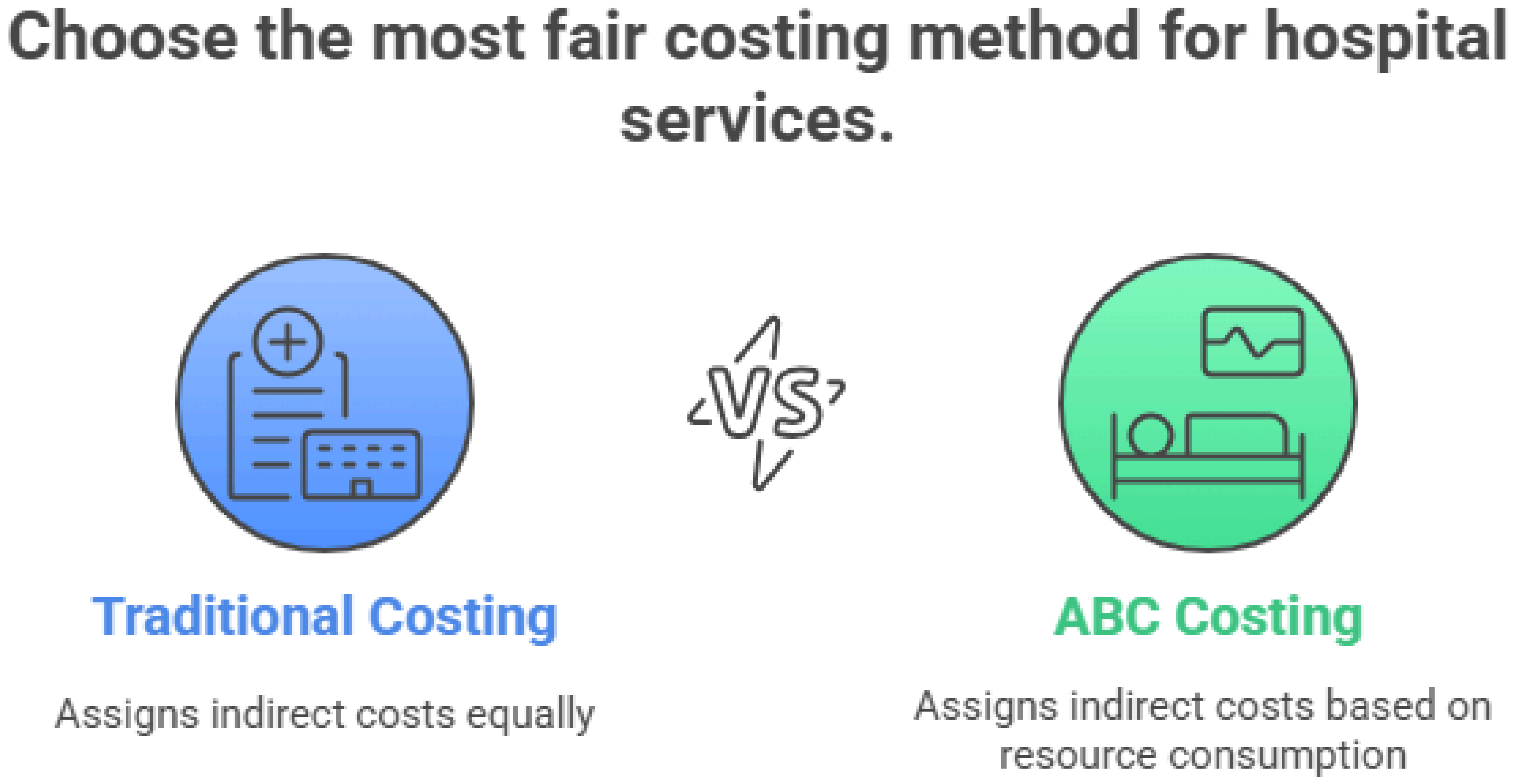
Why is this important?

It shows where overhead pressure is highest, helps management focus on improving processes in the most expensive area, and supports better pricing and budgeting for inpatient services.

Can you explain with an example?

Think of overheads like the "hidden costs" of running the hospital. If most of these "hidden costs" are used for inpatients, then improving efficiency in inpatient processes will give the biggest impact on total costs.


Which activity should be targeted to reduce costs?
Inpatient Admissions
Highest total cost and indirect cost absorption
Other Activities
Lower cost and indirect cost absorption


Choose the most fair costing method for hospital services.
VS
Traditional Costing
Assigns indirect costs equally
ABC Costing
Assigns indirect costs based on resource consumption

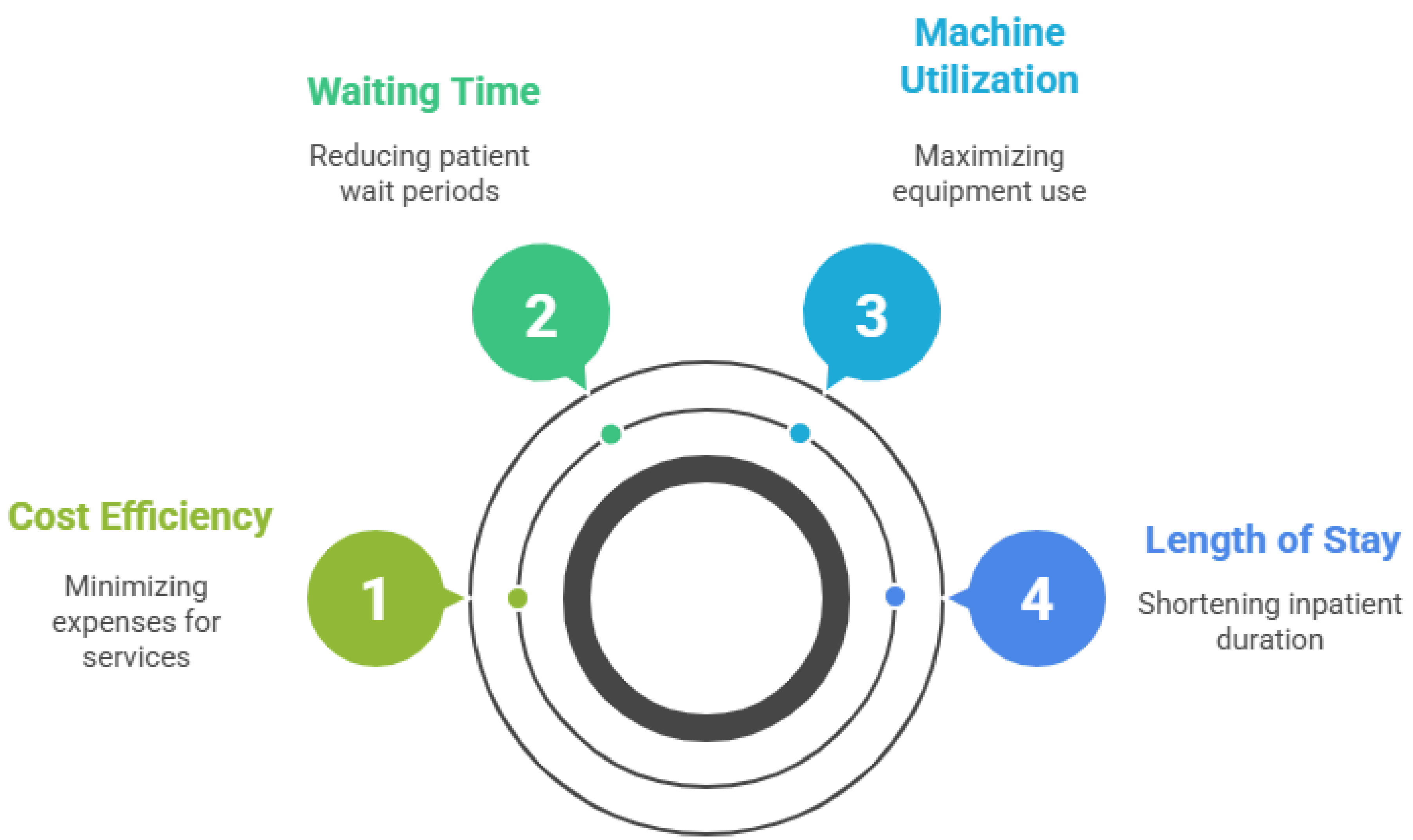
Efficiency Improvement Strategies

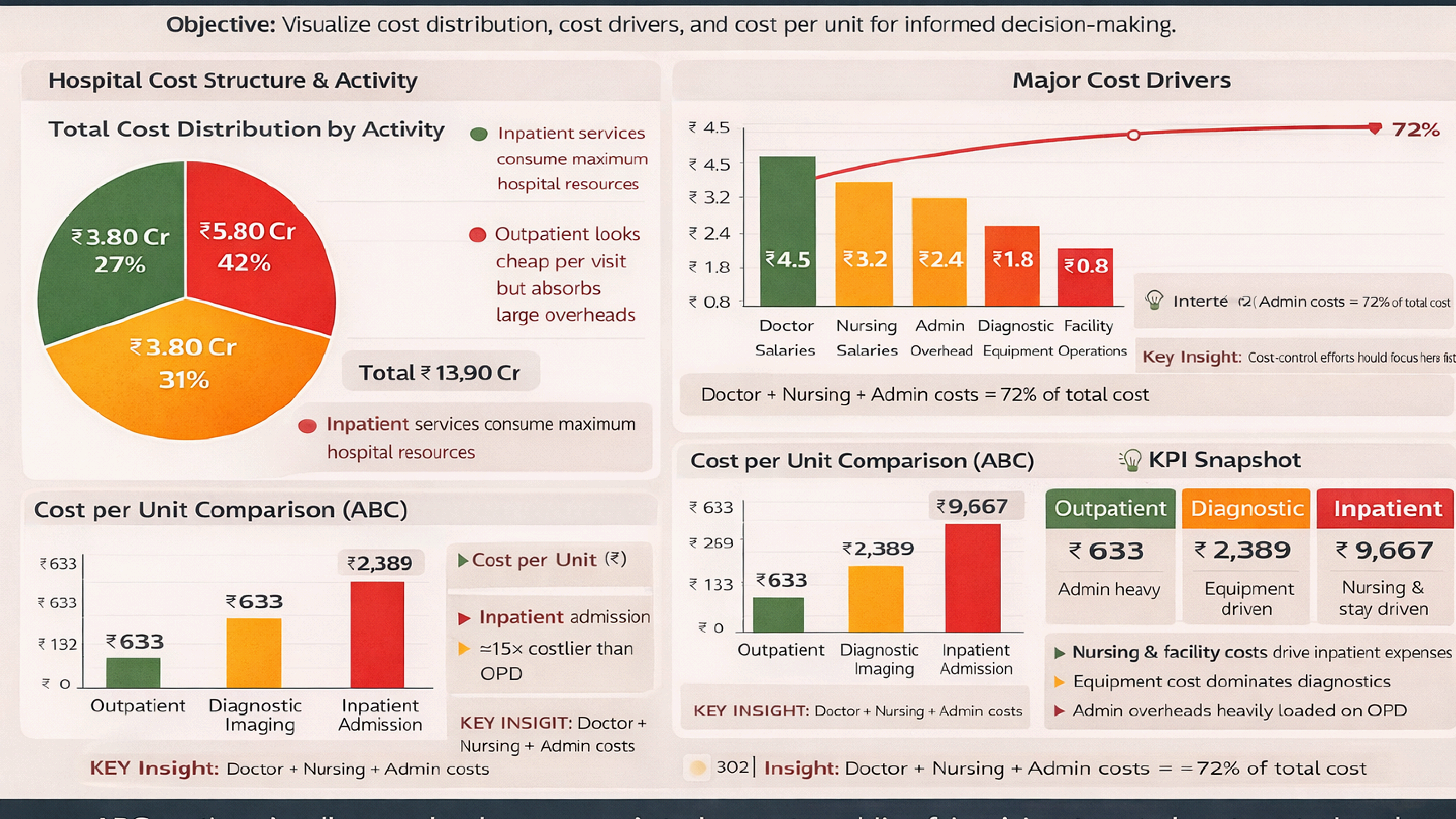
Strategy	Scheduling	Staffing	Process
Outpatient	Appointment automation	Task shifting to nurses	Reduce admin paperwork
Diagnostic Imaging	Batch scheduling of scans	Increase machine utilization	Preventive maintenance
Inpatient	Reduce average length of stay	Nurse workload balancing	Standard treatment protocols

Pricing Adjustments Based on ABC

Adjustment	Inpatient Package	Advanced Diagnostics	Outpatient Services
Description	Increase pricing	Differential pricing	Keep affordable, reduce admin

Hospital Performance Indicators




Objective: Visualize cost distribution, cost drivers, and cost per unit for informed decision-making.
Hospital Cost Structure & Activity
Total Cost Distribution by Activity
Inpatient services consume maximum hospital resources
₹3.80 Cr 27%
₹5.80 Cr 42%
₹3.80 Cr 31%
Outpatient looks cheap per visit but absorbs large overheads
Total ₹ 13,90 Cr
Inpatient services consume maximum hospital resources
Major Cost Drivers
₹ 4.5
₹ 4.5
₹ 3.2
₹ 2.4
₹ 1.8
₹ 0.8
72%
₹4.5
₹3.2
₹2.4
₹1.8
₹0.8
Doctor Salaries
Nursing Salaries
Admin Overhead
Diagnostic Equipment
Facility Operations
Interté (Admin costs = 72% of total cost
Key Insight: Cost-control efforts hould focus hers fist
Doctor + Nursing + Admin costs = 72% of total cost
Cost per Unit Comparison (ABC)
₹633
₹633
₹132
₹ 0
₹633
₹633
₹2,389
Outpatient
Diagnostic Imaging
Inpatient Admission
Cost per Unit (₹)
Inpatient admission
≈15× costlier than OPD
KEY INSIGIT: Doctor + Nursing + Admin costs
KEY Insight: Doctor + Nursing + Admin costs
Cost per Unit Comparison (ABC)
₹ 633
₹ 269
₹ 133
₹ 0
₹633
₹2,389
₹9,667
Outpatient
Diagnostic Imaging
Inpatient Admission
KEY INSIGHT: Doctor + Nursing + Admin costs
KPI Snapshot
Outpatient
₹ 633
Admin heavy
Diagnostic
₹ 2,389
Equipment driven
Inpatient
₹ 9,667
Nursing & stay driven
Nursing & facility costs drive inpatient expenses
Equipment cost dominates diagnostics
Admin overheads heavily loaded on OPD
302 | Insight: Doctor + Nursing + Admin costs = =72% of total cost

Activity - 2

Objective - To compare cost efficiency and cost drivers across three healthcare facilities (A, B, C) using a statistical approach to costing.

Facility Comparison

Parameter	Facility A	Facility B	Facility C
Number of Beds	100	250	500
Annual Case Load	12,000	40,000	85,000
Case Mix Complexity (1–10)	3	6	9
Case Severity Index (1–10)	2	5	8
Occupancy Rate (%)	60%	80%	92%
Maturity Level (1–5)	2	3	4

Cost Comparison of Facilities

Cost Measure	Facility A	Facility B	Facility C
Total Annual Cost (₹ Crore)	48	120	330
Average Cost per Patient (₹)	4,000	3,000	3,880
Average Cost per Bed per Year (₹ Lakh)	48	48	66
Average Cost per Patient-Day (₹)	2,750	2,100	2,950

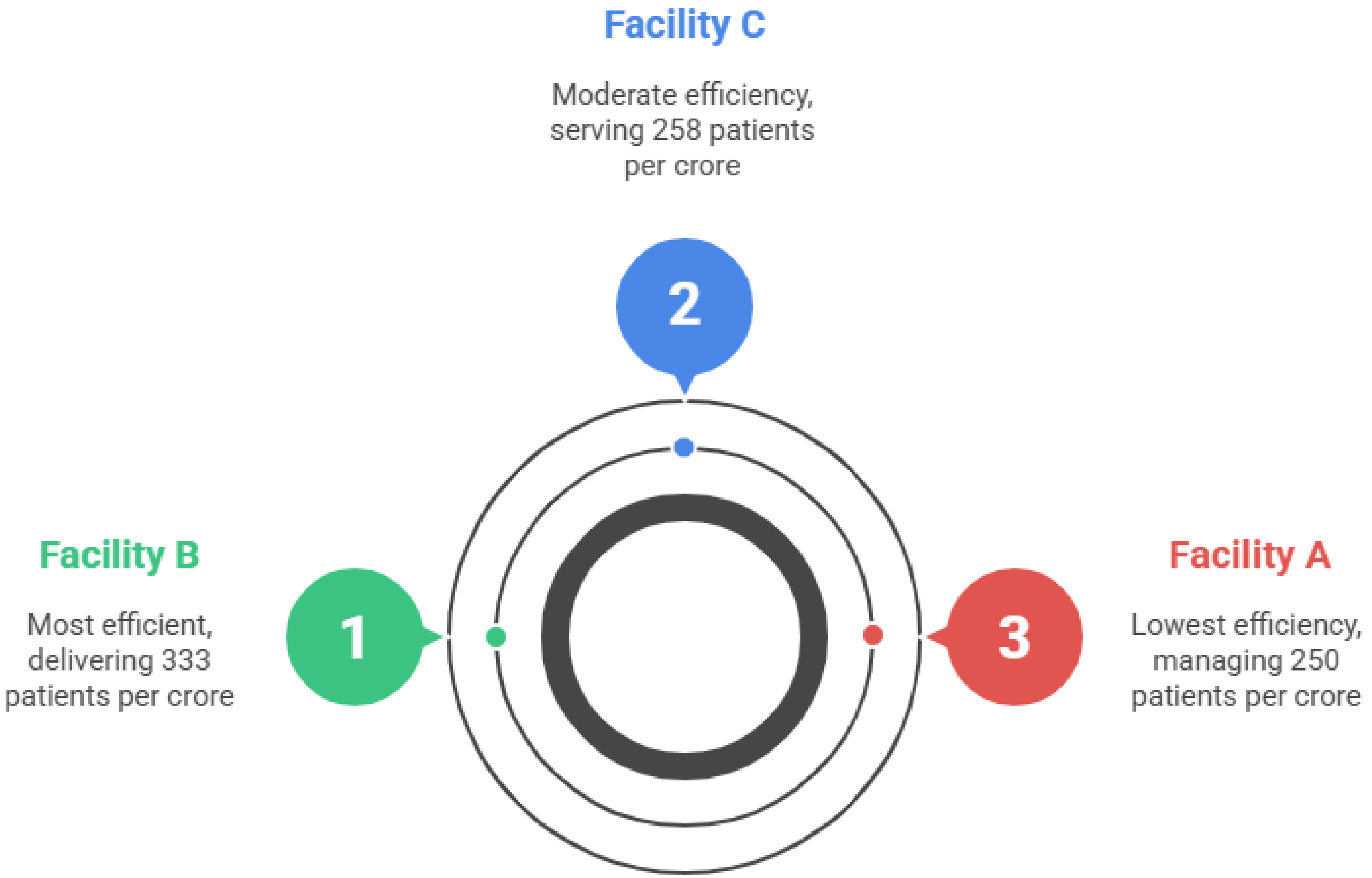
Cost Efficiency of three facilities

Efficiency = Patients ÷ Total Cost

Facility Comparison

Characteristic	Patients	Total Cost (₹ Cr)	Patients per ₹ Cr
Facility A	12,000	48	250
Facility B	40,000	120	333
Facility C	85,000	330	258

Hospital Efficiency Comparison



Bed Size Related to Cost Per Patient

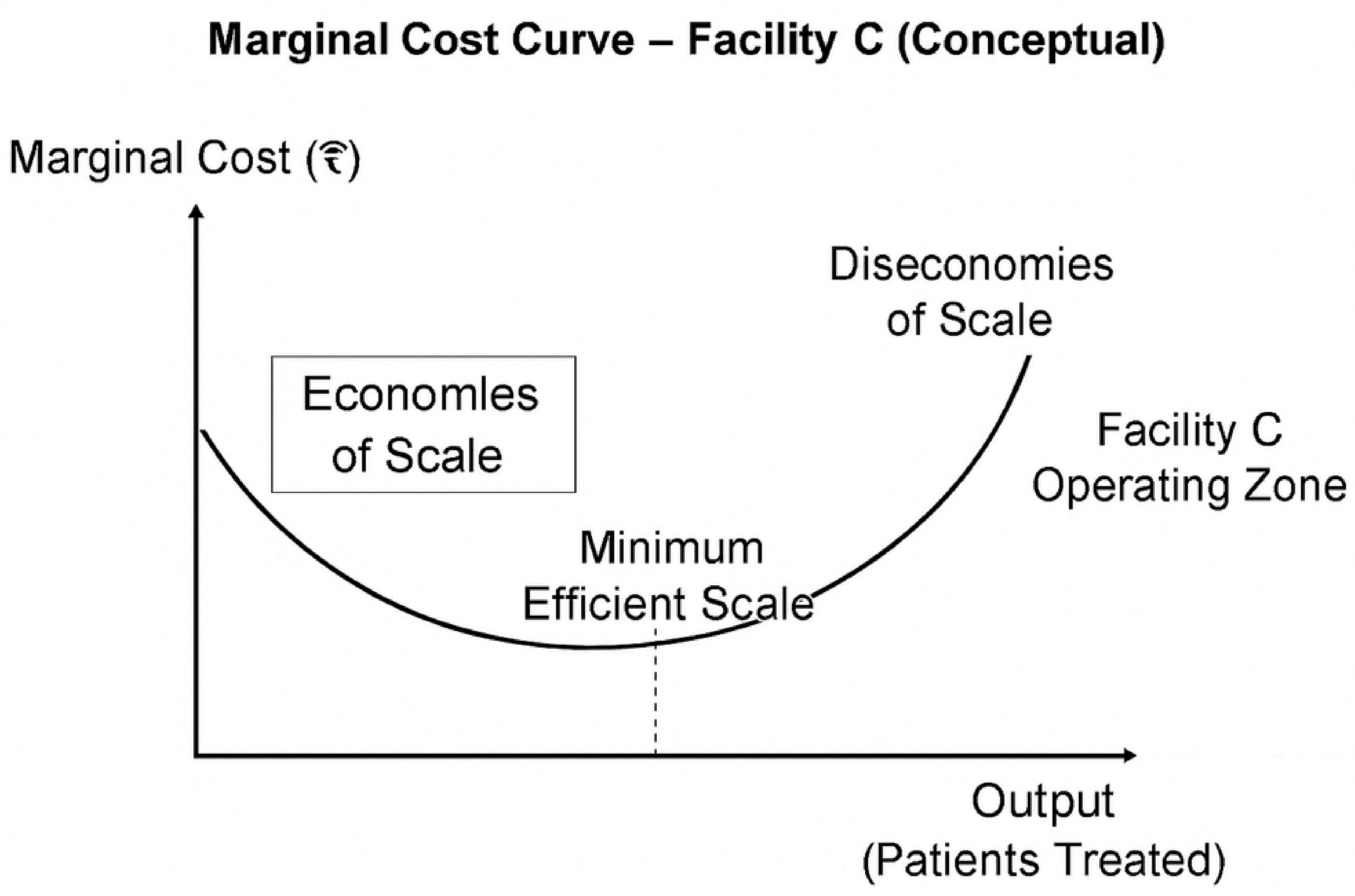
Bed Size: A = 100 | B = 250 | C = 500 Cost per
Patient: 4,000 → 3,000 → 3,880

Interpretation:

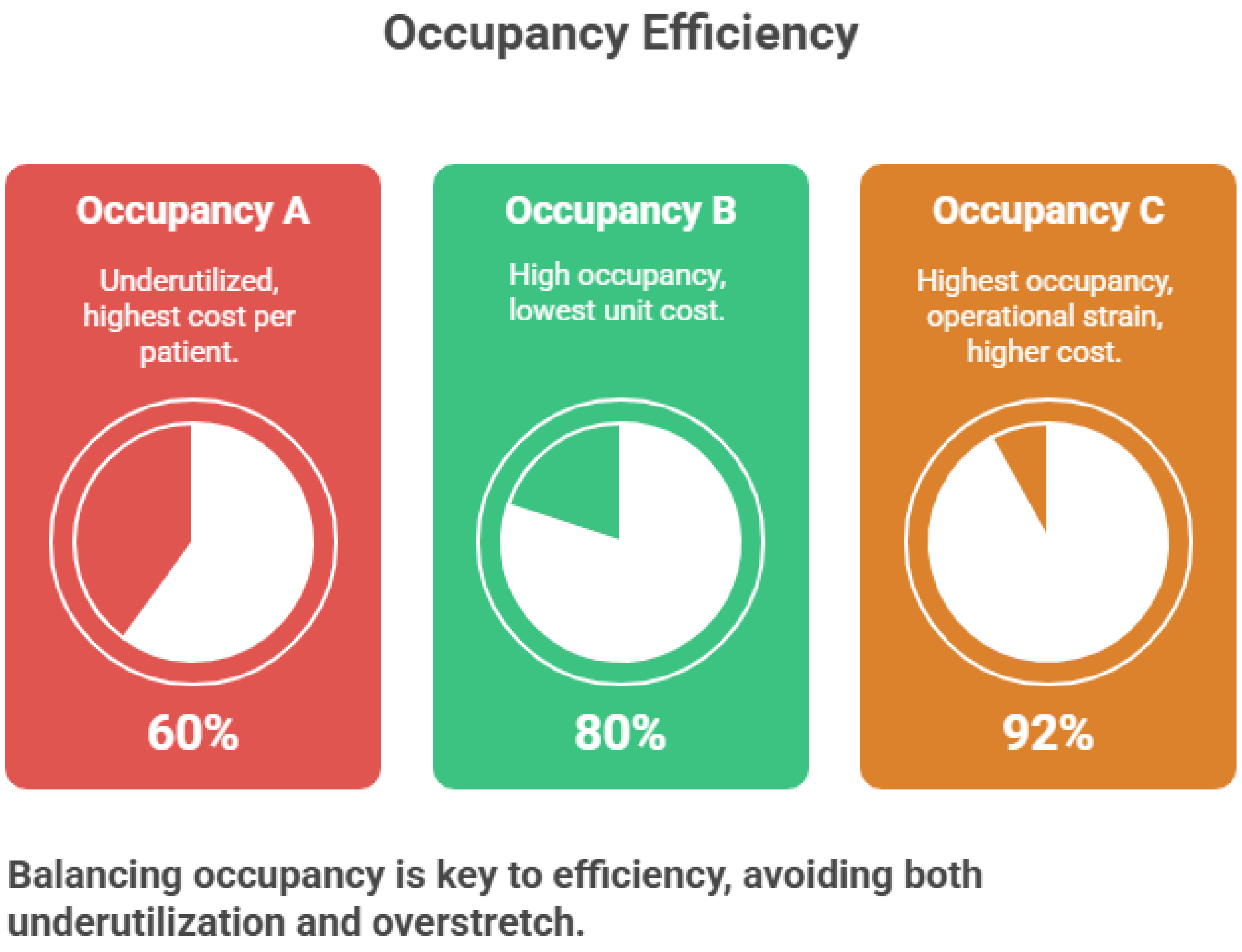
- From A → B: Cost per patient drops 25%, showing economies of scale.
- From B → C: Cost rises again (3,000 → 3,880) despite 2× bed size.

Why scale breaks at C:

- Case mix = 9 (highest)
- Severity = 8
- Case load = 85,000
- These increase marginal cost per patient, offsetting scale benefits.



Role of Occupancy & Bed Utilization



Occupancy: A = 60% | B = 80% | C = 92% (Highest)

Efficiency Interpretation:

- C uses beds most efficiently, but high occupancy + high severity → operational strain → higher cost per patient.
- B hits the "sweet spot": high occupancy without overstretch → lowest unit cost.
- A underutilized (60% occupancy) → fixed cost spread over fewer patients → highest cost per patient.

Cost Drivers – What Explains Variation?

Case Load

Higher load spreads fixed cost → lowers unit cost (A→B). But excessive load increases marginal cost (C).

Case Mix Complexity

Higher complexity (A=3, B6, C=9) → more diagnostics, specialist time → higher cost.

Case Severity

Severity (A=2, B=5, C=8) strongly correlates with resource intensity → increases cost per patient.

Maturity Level

Higher maturity (A=2, B=3, C=4) → better systems, protocols → cost efficiency improves, but only up to a point.

Occupancy Rate

Low occupancy → high cost per patient (A).
Optimal occupancy → best efficiency (B).
Very high occupancy → strain + cost e escalation (C).

Statistical Reasoning — Cost Correlations

Case Mix Complexity ↑ → Cost per Patient ↑

A = 3 → ₹4,000
B = 6 → ₹3,000
C = 9 → ₹3,880

Non-linear: B is efficient due to optimal occupancy and maturity.

Overall: Positive correlation, but moderated by other factors.

Occupancy ↑ → Cost per Patient ↓ (up to a point)

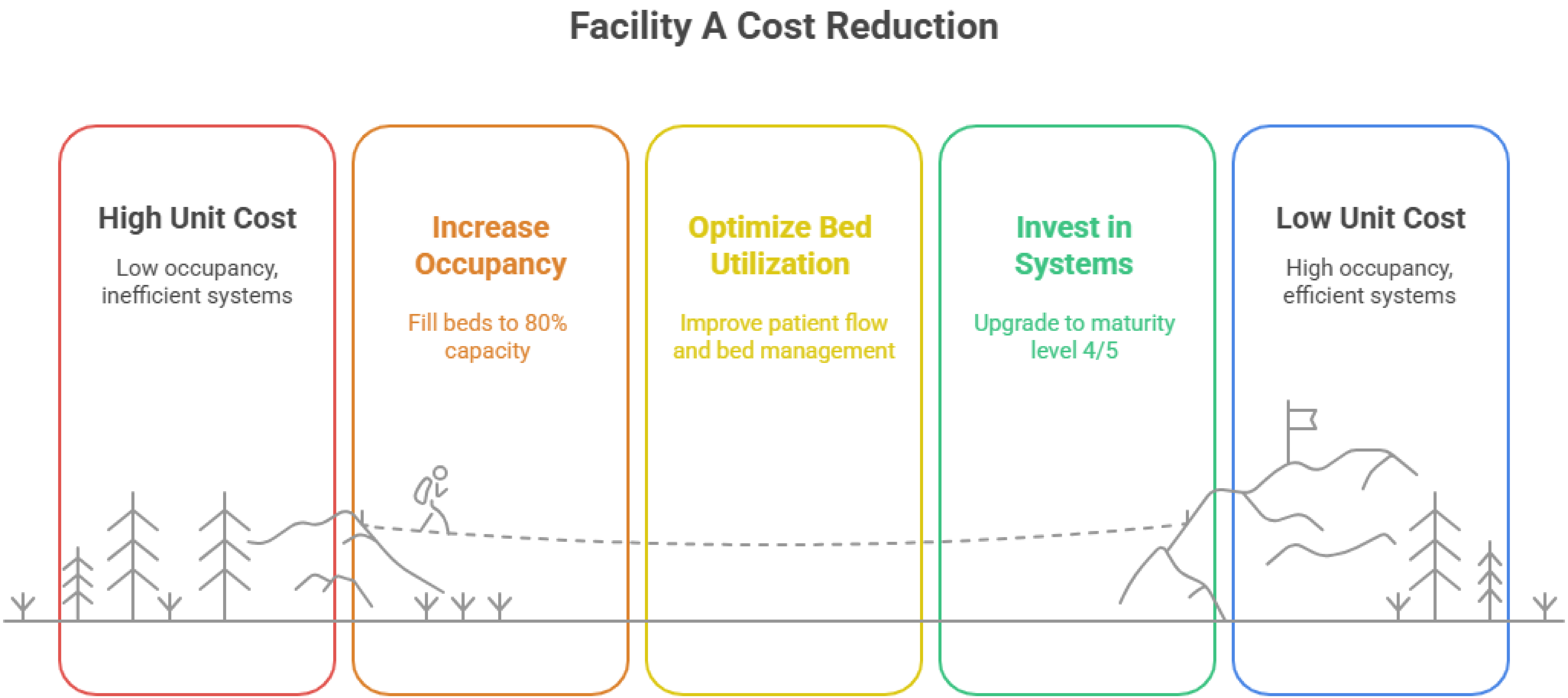
A = 60% → ₹4,000
B = 80% → ₹3,000
C = 92% → ₹3,880

Inverted U-shape: Efficiency improves till optimal occupancy, then cost rises due to strain.

Bed Size ↑ → Cost per Patient ↓ (not always)

A = 100 beds → ₹4,000
B = 250 beds → ₹3,000
C = 500 beds → ₹3,880

Economies of scale from A→B, but diseconomies at C due to overload.


Facility A Cost Reduction
High Unit Cost
Low occupancy, inefficient systems
Increase Occupancy
Fill beds to 80% capacity
Optimize Bed Utilization
Improve patient flow and bed management
Invest in Systems
Upgrade to maturity level 4/5
Low Unit Cost
High occupancy, efficient systems

Which facility appears "overstretched"?

Patient burden per bed (per year)

Patient burden per bed=Annual case load/Number of beds

- **Facility A:**

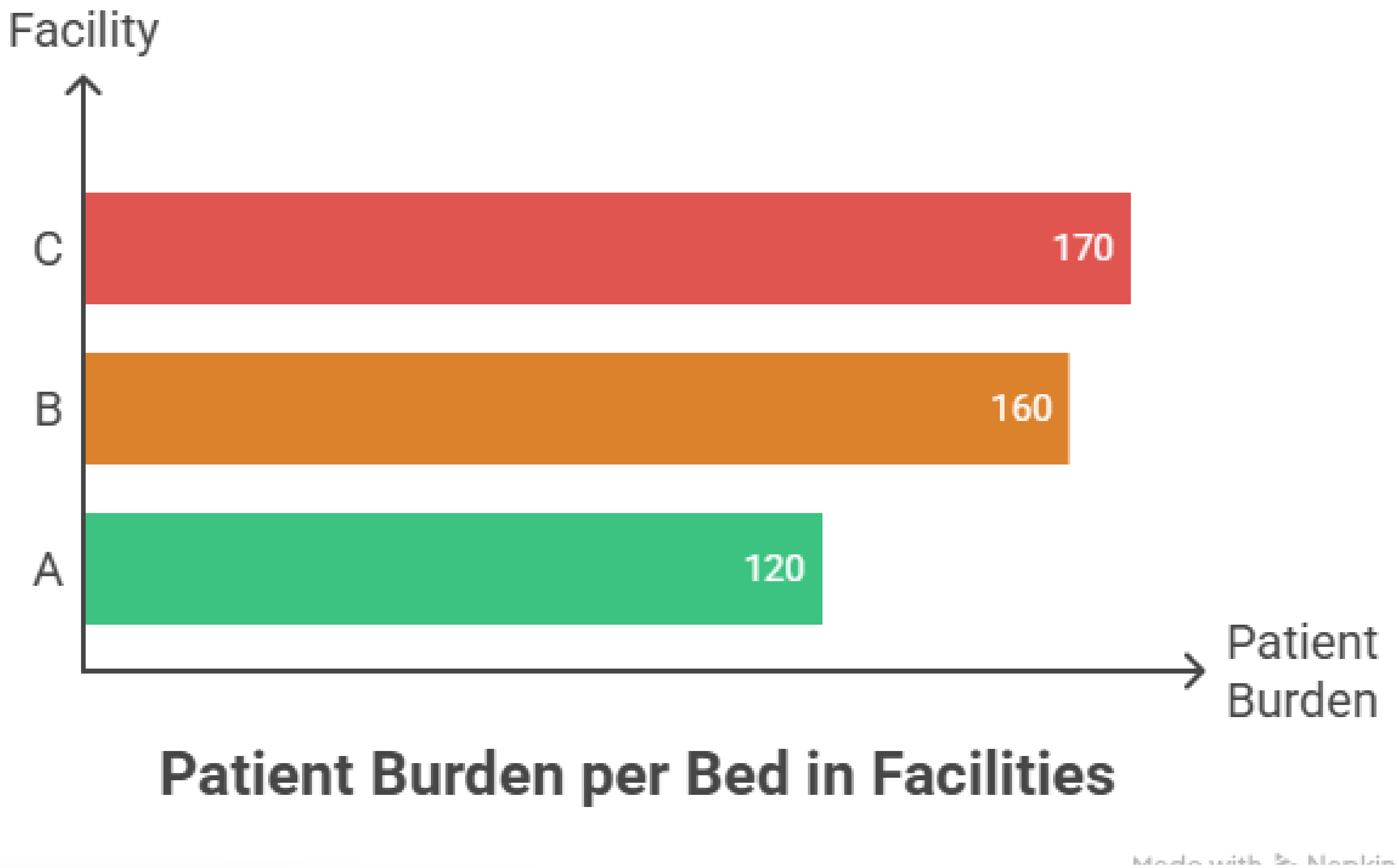
12,000/100=120 patients per bed per year

- **Facility B:**

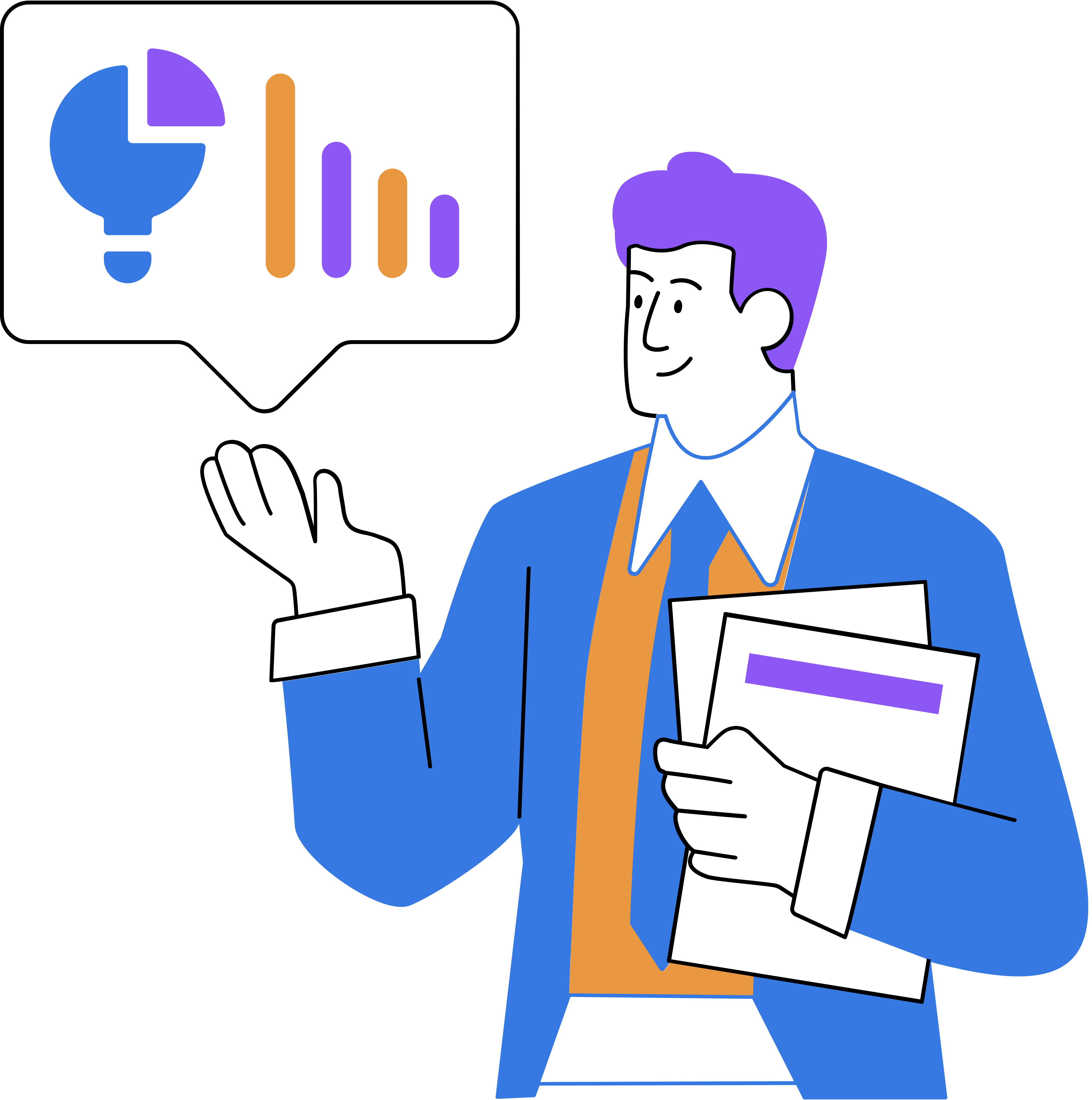
40,000/250=160 patients per bed per year

- **Facility C:**

85,000/500=170 patients per bed per year



This increases risks of staff burnout, quality compromise, rising marginal cost, and patient dissatisfaction."



Thank You

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